

StayPG

Verified PG / Hostel Discovery & Booking Platform for Students in Patiala

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What is StayPG?

An Airbnb-style MERN marketplace built for Patiala's student housing market. Verified owners list PG/hostel properties with photos, pricing and amenities; students search, filter, locate on a map, chat with owners, book a stay and leave reviews — all secured by JWT-based, role-aware authentication.

Project Objective

Design and develop a secure, full-stack MERN web application that connects PG/hostel owners in Patiala with students through verified property listings, map-based search and filtering, real-time in-app chat, a booking workflow and a post-stay review and rating system.

Motivation: Why Build StayPG?

A real, local housing gap for students in Patiala, tackled with production-style full-stack engineering.

1 An Urgent Local Need

Every year, blind and rushed decisions

- Thousands of students relocate to Patiala each year needing a PG/hostel within their first few weeks.
- Search today runs on word-of-mouth, local agents and scattered WhatsApp/Facebook posts.
- Rushed, unverifiable decisions expose students to overpriced rent and fraudulent listings.

2 Two-Sided Marketplace Gap

Airbnb-style trust, built for PGs

- Genuine owners have no dedicated digital channel and stay dependent on brokers.
- A verified-listing marketplace builds trust for students and reach for owners.
- Reviews, ratings and a 'Verified Owner' badge create accountability on both sides.

3 Full-Stack Engineering Practice

MERN + real-time + geolocation

- Combines REST APIs, JWT auth, geolocation and Socket.io real-time chat in one system.
- Goes beyond CRUD with role-based access control across Student, Owner and Admin.
- A strong, practical demonstration of production-style full-stack architecture.

Problem Statement: The PG Housing Gap

Students and genuine owners both lose out in a fragmented, offline and unverifiable market.

1

Unverified, Fragmented Search

No single trustworthy source

- No platform to compare verified listings, real photos or genuine tenant reviews.
- Students depend on informal WhatsApp/Facebook groups and local property agents.
- Comparing price, distance and amenities across scattered sources is slow and manual.

2

Information Asymmetry & Fraud

High-stakes decisions, low visibility

- New students can't verify a landlord's identity or a room's real condition remotely.
- Misrepresented rooms and overpriced rent are common for first-time renters.
- No accountability system exists today to flag or filter out bad actors.

3

No Digital Channel for Owners

Broker-dependent, low occupancy

- Genuine owners have no way to showcase properties or manage bookings online.
- Dependence on brokers limits occupancy, revenue and reach for good listings.
- No structured booking/request workflow — everything is manual and ad hoc.

Proposed Solution: StayPG System Architecture

A 3-tier MERN architecture with role-based auth, real-time chat and map-based search.

1 Frontend Dashboard

React.js + Redux/Context + Tailwind

- SPA with separate Student, Owner and Admin dashboards.
- Search & filter UI with interactive map view of nearby properties.
- Live chat window, booking status tracker and review forms.

2 Backend, Auth & Real-Time

Node.js + Express + JWT + Socket.io

- REST APIs for listings, bookings, reviews and admin verification.
- JWT access tokens + HTTP-only refresh cookies; bcrypt password hashing.
- Role-based middleware restricting routes for Student / Owner / Admin.

3 Data & Integrations

MongoDB + Cloudinary + Google Maps

- MongoDB/Mongoose schemas for users, properties, bookings and chats.
- Multer + Cloudinary for multi-image property uploads.
- Google Maps Geocoding API for distance-from-college search.

PROJECT ROADMAP & MODULE PHASES

Phase 1: Auth & Role Management → Phase 2: Listings, Search & Maps → Phase 3: Booking & Reviews → Phase 4: Real-Time Chat & Deployment

Core Feature Modules

Six modules covering both sides of the marketplace, from listing to post-stay review.

1 User & Role Management

Student / Owner / Admin

- Registration & login for three roles with separate dashboards.
- Profile management incl. verification details.

2 Property Listing (Owner)

CRUD + media

- Multi-image upload via Multer & Cloudinary.
- Rent, amenities, sharing type and vacancy toggle.

3 Search, Filter & Locate

Student-side discovery

- Filter by price, gender preference, distance and rating.
- Interactive map showing distance from selected college.

4 Booking Workflow

Request-based scheduling

- Request → Accept / Reject / Reschedule flow.
- Status tracking: Pending, Confirmed, Cancelled, Completed.

5 Review & Verification

Trust & accountability

- Reviews allowed only after a completed booking.
- Admin verifies owner documents before listings go live.

6 Real-Time Chat

Socket.io messaging

- Live one-to-one chat between student and owner.
- Full chat history stored in MongoDB.

12-Week Roadmap, Validation & Budget

An iterative build-and-test plan executed on a zero-cost, free-tier technology stack.

12-Week Delivery Timeline

- **Weeks 1–2**
Planning — requirement gathering, DB schema design, wireframes.
- **Weeks 3–6**
Backend — auth/role middleware, property CRUD, search & filter APIs.
- **Weeks 7–9**
Frontend & core flows — React UI, booking workflow, review module.
- **Weeks 10–12**
Chat, maps, testing & deployment — Socket.io, Google Maps, cloud deploy.

Testing & Validation

- Unit Testing (Jest) — booking logic, rating calculation
- API Testing (Postman) — incl. unauthorized-access cases
- Integration Testing — signup → listing → booking → review
- Security Testing — JWT tampering, role-escalation checks
- UAT — real students & PG owners in Patiala trial the app

Estimated Budget

Rs. 0 – Rs. 800

All core resources on free tiers — MongoDB Atlas, Cloudinary, Render/Vercel, Google Maps free credit. Domain (optional) ≈ Rs. 800/yr.

The Gap. The Solution. The Impact.

Elevator pitch summary for StayPG.

1 The Gap

No trust, no single source

- Students moving to Patiala have no verified, centralised platform to find trustworthy PG/hostel accommodation.

2 The Solution

StayPG marketplace

- A MERN-stack marketplace where verified owners list properties and students search, locate, chat, book and review — behind secure, role-based authentication.

3 The Impact

Safer housing, real reach

- A safer, more transparent way for Patiala's student community to find housing — and a proper digital channel for genuine PG owners to reach verified tenants.

StayPG — Thank you.